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Open Macroeconomics - Final Exam

A Replication of Uribe and Yue (2006), with an Extension to Brazil, 1996–2026

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This exercise addresses interest rate shocks in emerging market economies. It is a formal replication of the small open economy model of Chapter 6 of Schmitt-Grohé and Uribe's (2017) *Open Economy Macroeconomics*, itself based on Uribe and Yue (2006), together with an original data extension to Brazil, sampled 1996:Q1–2026:Q1.

1. Introduction

Motivation: Business cycles in emerging market economies move closely with the interest rates these countries face in international financial markets.

- Low country interest rates $\leftrightarrow$ economic expansions.
- High country interest rates $\leftrightarrow$ depressed aggregate activity.

What fraction of business-cycle fluctuations in emerging markets is due to movements in the country interest rate?
Answer is non trivial. The country interest rate is not exogenous to the country's own domestic conditions. Spreads respond systematically to output, investment, the trade balance — reverse causality is a first-order concern.

The identification strategy: To quantify the macroeconomic effects of interest-rate shocks, the first step is to isolate the exogenous component of country spreads. Following Uribe and Yue (2006), identification will combine two ingredients:

- Statistical methods — to identify shocks from the data.
- Economic theory — to assess whether the *effects* of those identified shocks behave the way a genuine structural shock's effects should: in practice, whether the estimated responses to the identified shocks resemble what theoretical model predicts.

2. Methodology

- **Identify country-spread and US-interest-rate shocks:** using a VAR with a set of restrictions on the system's coefficient matrices (the recursive ordering). The imposed restrictions, nonetheless, cannot be tested using the VAR data alone;
- **Hypothesize a theory of the business cycle:** an external benchmark is needed to judge whether those restrictions, and hence the identified shocks, are credible.
- **The approach:** impose the shocks found empirically (VAR system) to a theoretical (DSGE) model, then compare the *resulting business-cycle dynamics* (IRF´s).
- **Verify the model validity:** If the two systems generate similar fluctuations in output, investment, and the trade balance under a common shock, this supports treating the empirically identified shocks as genuine structural disturbances of the kind the theory has in mind **and** we can conclude the model is valid.

Algorithm:

1. Estimate a VAR system linking the world rate, the country rate, and domestic variables; identify shocks via a recursive restriction.

2. Feed the VAR-estimated country and world interest rate (R and R^{us}) processes into the DSGE model as its exogenous driving forces.

3. Estimate the DSGE's structural frictions by IRF matching (in the style of Christiano, Eichenbaum, and Evans (2001)): choose parameters so the model's theoretical impulse responses best match the VAR's estimated ones.

4. Impose a common shock in both systems and compare the resulting business-cycle dynamics (impulse responses) — VAR-implied vs. DSGE-implied.

Conclusion logic: Similar business-cycle dynamics under a common shock $\Rightarrow$ the identified shocks are plausible. Large, systematic divergence $\Rightarrow$ the identification, the model, or the estimation sample warrants scrutiny.

3. An Empirical Model

- A 5-variable VAR system linking domestic and external variables:
  - y_t — real output (log-deviation from a log-linear trend)
  - i_t — real investment (log-deviation from a log-linear trend)
  - tby_t — trade balance-to-output ratio
  - R_t^{us} — gross world (US) real interest rate (log)
  - R_t — gross country real interest rate (log)
- Output, investment, and the trade balance are seasonally adjusted.
- Full variable construction and data sources: Uribe and Yue (2006), Section 2.1.

Restrictions and outputs:

- Recursive (Cholesky) identification: the contemporaneous matrix A is lower-triangular with unit diagonal.
  - Real variables (y, i, tby) react to spread / world-rate innovations only with a one-quarter lag.
  - Financial variables (R^{us}, R) react to real shocks contemporaneously.

- R_t^{us} is restricted to a univariate AR(1) process — a small emerging economy cannot move the US interest rate.

- Our replication code reproduces Table 6.1 (parameter estimates) and the book's own impulse response figures.

Files: original data — `statadata.xlsx` (baseline 7-country panel), `irs_data_update.mat` (the textbook's own expanded 16-country, 1994–2012 robustness-check panel, Ch. 6, Section 6.2.1); replication code (MATLAB) — `uribe_yue_VAR_baseline_v2.mlx`.

4. The DSGE Model: An Open Economy Subject to Interest-Rate Shocks

- Standard small open economy neoclassical growth model (Mendoza 1991), extended with four key frictions:

- **1. Time-to-build capital:** Investment projects take 4 quarters to complete, plus convex capital-adjustment costs ⇒ smooth, hump-shaped investment dynamics;

- **2. External habit formation** ("catching up with the Joneses") ⇒ prevents excessive consumption volatility;

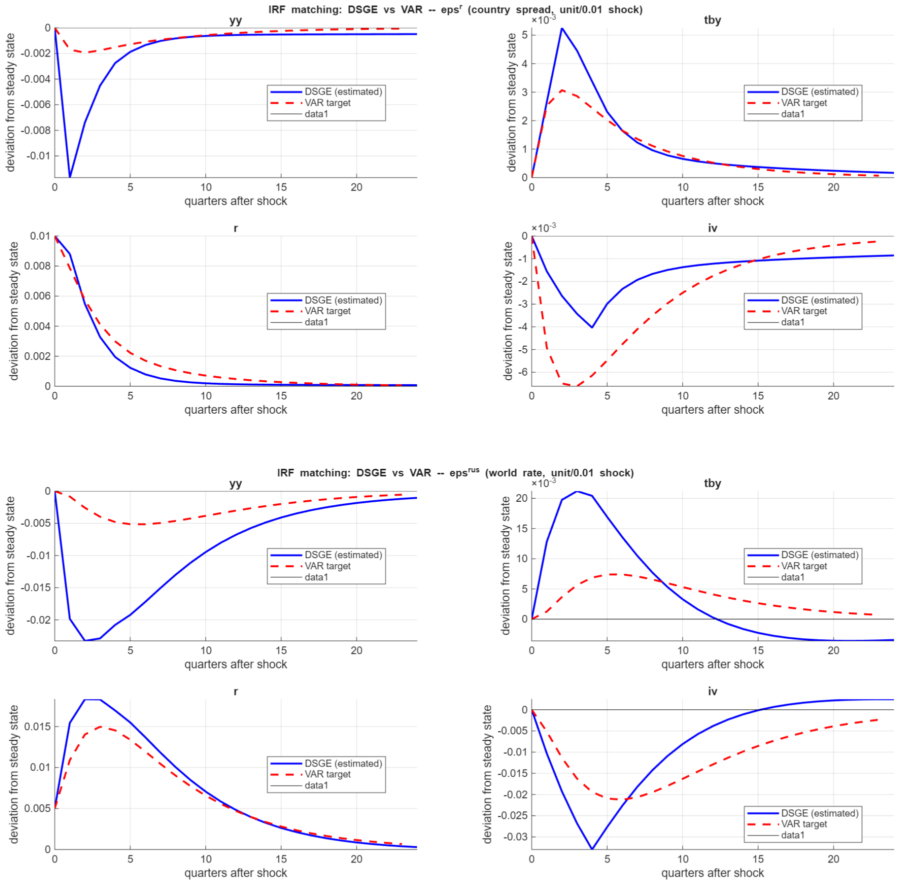
- **3. Working-capital-in-advance constraint:** Firms must finance a fraction η of the wage bill in advance ⇒ a rate hike raises firms' effective labor cost directly, producing a genuine supply-side contraction;

- **4. Information timing:** Production and absorption decisions are made one period before that period's interest-rate shock is realized — mirrors the VAR's own identifying assumption exactly.

Replication of the baseline model:

- **Model specification:** Full equations and derivations can be found in Uribe and Yue (2006), Schmitt-Grohé and Uribe (2017) (Ch. 6), and our own derivation document (`uribe_yue_derivation`) — which further translates every resulting equation into Dynare code (`DSGE_uribe_yue_IRFmatch.mod`).

- **Results:** The two sets of business-cycle dynamics — VAR-implied and DSGE-implied, both under the same shock — are close in this baseline case, confirming findings of the original study (Uribe and Yue 2006).



Baseline (7-country panel) IRF comparison — VAR-estimated vs. DSGE-implied, country-spread and world-rate shocks.

- **Conclusion:** this supports treating the identified country-spread and US-rate shocks as economically plausible — the theory, taken seriously, reproduces the dynamic effects the data-driven VAR attributes to these shocks.

5. Extending the Model to Brazil Only, 1996–2026

While panel data provides larger effective sample and more efficient estimates, it also imposes a homogeneity assumption on the *dynamic* (slope) coefficients: Argentina, Brazil, Ecuador, etc are assumed to share the same estimated responses to shocks (even though country-specific intercepts are accounted for via fixed effects). A single country sample over a much longer horizon relaxes cross-country homogeneity entirely, but now requires coefficient stability over a much longer, more heterogeneous period — difficult to defend across the GFC, COVID, and the 2021–23 hiking cycle. Idiosyncratic events affecting Brazil during this period (e.g., political turmoil) are also another likely source of regime changes.

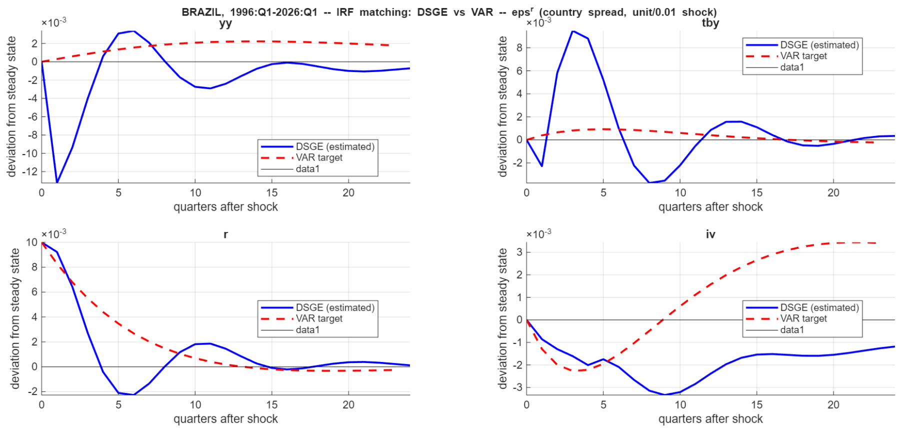
Files: `uribe_yue_VAR_Brazil.mlx` (Brazil-only VAR replication) — `DSGE_uribe_yue_IRFmatch_Brazil.mod` (Dynare model, Brazil-specific R/R^{us} processes) — `replicate_IRF_uribe_yue_matching_Brazil.m` (DSGE estimation + IRF comparison).

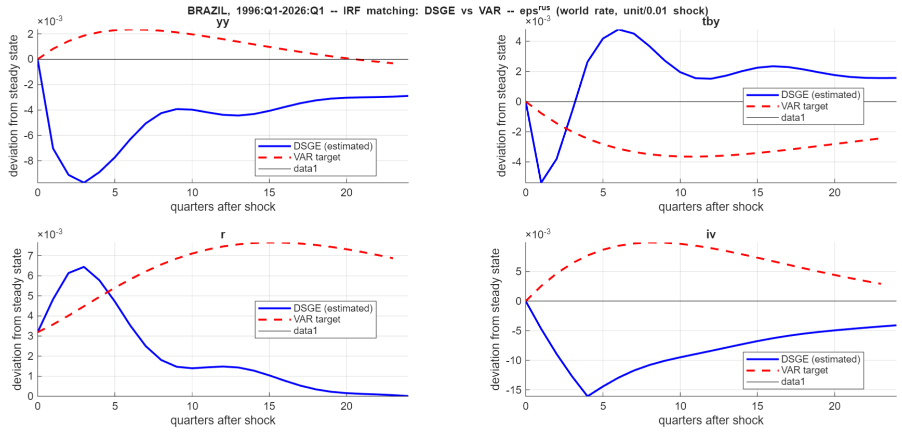
6. Analysing Results for Brazil

Preliminary considerations: Many caveats apply before treating these results as conclusive.

- **The model — calibrated as in Uribe and Yue (2006) — does not validate well when looking at Brazil-only data**, a signal that is masked when pooled into the 7-country panel;
- **Higher persistence of R^{us}**: In the original paper, $\rho \approx 0.83$. Once extended to 2026 — which now spans the post-GFC near-zero-rate — persitence is ≈ 0.96, which dramatically slows the estimated IRFs' convergence back to steady state;
- This pattern actually confirms one of the robustness checks in Schmitt-Grohé and Uribe (2017): Section 6.2.1 extended the sample through 2012, and likewise found *significantly more persistent* estimated responses than the original 1994–2001 baseline.

IRF comparison:





Brazil-only IRF comparison — VAR-estimated vs. DSGE-implied, country-spread and world-rate shocks.

- **VAR and DSGE business-cycle dynamics diverge sharply** — especially the response of output, which stays economically implausible even after re-estimating the model's frictions.
- We also tested relaxing the VAR's "same-quarter exogeneity" identifying assumption; the divergence got worse, not better — by the paper's own logic, this argues against identification being the culprit.
- **More likely explanation:** Omitted common drivers (e.g. commodity prices / terms of trade) may move spreads and output together, contaminating the identified "shocks"
- **Takeaway:** a genuine, informative external-validity failure — this model, calibrated as in Uribe and Yue (2006), does not straightforwardly extend to Brazil over this long, heterogeneous sample.

References

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